

Bioavailability and inter-conversion of sulforaphane and erucin in human subjects consuming broccoli sprouts or broccoli supplement in a cross-over study design

Broccoli consumption may reduce the risk of various cancers and many broccoli supplements are now available. The bioavailability and excretion of the mercapturic acid pathway metabolites isothiocyanates after human consumption of broccoli supplements has not been tested. Two important isothiocyanates from broccoli are sulforaphane and erucin. We employed a cross-over study design in which 12 subjects consumed 40 grams of fresh broccoli sprouts followed by a 1 month washout period and then the same 12 subjects consumed 6 pills of a broccoli supplement. As negative controls for isothiocyanate consumption four additional subjects consumed alfalfa sprouts during the first phase and placebo pills during the second. Blood and urine samples were collected for 48 hours during each phase and analyzed for sulforaphane and erucin metabolites using LC-MS/MS. The bioavailability of sulforaphane and erucin is dramatically lower when subjects consume broccoli supplements compared to fresh broccoli sprouts. The peaks in plasma concentrations and urinary excretion were also delayed when subjects consumed the broccoli supplement. GSTP1 polymorphisms did not affect the metabolism or excretion of sulforaphane or erucin. Sulforaphane and erucin are able to interconvert in vivo and this interconversion is consistent within each subject but variable between subjects. This study confirms that consumption of broccoli supplements devoid of myrosinase activity does not produce equivalent plasma concentrations of the bioactive isothiocyanate metabolites compared to broccoli sprouts. This has implications for people who consume the recommended serving size (1 pill) of a broccoli supplement and believe they are getting equivalent doses of isothiocyanates.